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RESEARCH ARTICLE

Physics-Guided Multimodal Thermography for Non-Invasive Core Temperature Estimation of Frozen Foods From Instantaneous Surface Thermal Observations

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ABSTRACT Accurate core temperature monitoring is essential for food safety and quality control in frozen food processing, yet conventional sensing methods are invasive and unsuitable for non-destructive in-line inspection. This paper presents a physics-guided multimodal thermography framework for estimating frozen-food core temperature from a single instantaneous surface thermal observation. Synchronized Red–Green–Blue (RGB) and Long-Wave InfraRed (LWIR) images are integrated through RGB-based YOLO26-seg segmentation, ROI-aware surface temperature extraction, and a heat-transfer-based estimation model operating under the regular regime. Experimental validation using 9,000 synchronized RGB–LWIR image pairs from 45 frozen cooked rice samples across three independent IQF batches demonstrated accurate estimation over a core-temperature range of −25 °C to −18 °C, achieving a Mean Absolute Error (MAE) of 0.4553 °C, a Root Mean Square Error (RMSE) of 0.5814 °C, and an R^2 of 0.9050 against probe-based measurements. The results demonstrate the feasibility of instantaneous, non-invasive core-temperature estimation without embedded sensing or continuous temporal tracking, providing a practical basis for in-line thermal monitoring of frozen foods. Beyond conventional compliance monitoring, the proposed system also supports the detection of both insufficient freezing and unnecessary overcooling, thereby contributing to improved food safety, thermal quality control, and energy-efficient frozen food processing.

INDEX TERMS Core temperature estimation, multimodal thermography, frozen food monitoring, non-invasive inspection, heat transfer modeling.

I. INTRODUCTION

Frozen food is essential in modern supply chains due to its convenience, prolonged shelf life, and preservation of product quality. However, frozen products are highly sensitive to temperature fluctuation, where even minor thermal deviations may accelerate microbial activity and compromise food safety. Consequently, strict temperature regulation is a fundamental requirement within Hazard Analysis and Critical Control Point (HACCP) systems [1]. Among thermal indicators, core temperature is particularly important because it directly reflects the internal thermal condition of the product, which cannot be reliably inferred from surface observation alone. Conventional core temperature measurement methods, including thermocouples, penetration probes, and embedded data loggers [2], [3], provide accurate measurements but remain invasive, destructive, and unsuitable for continuous in-line inspection [4]. These approaches are also difficult to scale for high-throughput industrial freezing environments, where only sparse sampling can practically be performed.

The challenge becomes more significant in industrial Individual Quick Freezing (IQF) systems, where products

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continuously move along conveyor lines with limited optical accessibility. Under such conditions, sensing systems are typically restricted to a single inspection checkpoint, rendering continuous thermal tracking and time-series-based estimation approaches impractical for real-world deployment. To overcome these limitations, this work proposes a snapshot-based multimodal thermography framework for non-invasive frozen food core temperature estimation using synchronized Red-Green-Blue (RGB) and Long-Wave InfraRed (LWIR) imaging. The proposed framework integrates synchronized RGB-LWIR thermography, ROI-aware thermal extraction, and established heat-transfer principles within the regular regime to estimate internal core temperature from a single synchronized RGB-LWIR image pair acquired at a designated inspection checkpoint, while maintaining low computational complexity suitable for industrial deployment. Although infrared thermography can readily measure surface temperature, the internal core temperature remains an inaccessible thermal state. Because substantial thermal gradients may exist between the surface and interior, the relationship between externally observable surface temperature and internal thermal condition is not inherently guaranteed. Consequently, the scientific question addressed in this work is whether the hidden core temperature can be reliably inferred from a single instantaneous surface thermal observation.

The major contributions can be summarized as follows:

1) Development of a multimodal RGB-LWIR thermography framework integrating segmentation-guided thermal ROI extraction with established heat-transfer principles within the regular regime, enabling non-invasive frozen-food core temperature estimation from a single instantaneous observation without continuous thermal tracking.
2) Comprehensive experimental validation using 45 frozen cooked rice samples from three independent IQF production batches, demonstrating the feasibility and consistency of the proposed physics-guided core temperature estimation framework across independent production batches.
3) Demonstration of the framework's practical applicability to CCP monitoring through non-invasive identification of both insufficient freezing and excessive overcooling conditions, supporting improved food safety, thermal quality control, and energy-efficient frozen food processing.

The remainder of this paper is organized as follows. Section II reviews the theoretical background and related studies relevant to this work. Section III presents the proposed methodology and system architecture. Section IV reports the experimental results and performance evaluation. Finally, Section V concludes the paper and discusses future research directions.

II. LITERATURE REVIEW

This section reviews the key concepts underlying the proposed framework, including HACCP-based temperature monitoring, heat transfer principles, multimodal thermography, computer vision-based segmentation, and existing non-invasive temperature estimation approaches.

A. HACCP AND CORE TEMPERATURE IN THE FOOD INDUSTRY

Hazard Analysis and Critical Control Point (HACCP) [1] is a widely adopted framework for ensuring food safety through controlled processing conditions. In frozen food systems, maintaining the product core temperature below $-18\,^{\circ}\mathrm{C}$ is essential to suppress microbial activity and preserve product quality [5], [6]. Since the internal thermal state cannot be directly observed, reliable core temperature monitoring is a critical requirement in industrial cold-chain operations.

Conventional measurement methods, including thermocouples and embedded data loggers [2], [3] as in Fig. 1, provide accurate internal temperature readings but remain invasive, destructive, and unsuitable for large-scale in-line inspection [4]. In addition, embedded sensors [7] often exhibit thermal inertia and limited placement flexibility, reducing sensitivity to rapid thermal variation. Although alternative non-invasive techniques such as ultrasonic sensing [8], Near-InfraRed (NIR) Spectroscopy [4], and Magnetic Resonance Imaging (MRI) and Nuclear Magnetic Resonance (NMR) [9] have been investigated, their industrial applicability remains limited by high complexity, restricted scalability, or insufficient robustness. These limitations motivate the development of practical non-invasive temperature estimation frameworks for real-time industrial deployment.



(a) Button Embedded Data Logger (b) Thermistor Data Logger

FIGURE 1. Two distinct types of data loggers.

B. HEAT TRANSFER PRINCIPLE

Heat transfer in solid systems is commonly analyzed using a One-Dimensional (1D) approximation when thermal variation is dominated along the depth direction and lateral gradients are negligible [10]. This assumption is widely adopted in slab-like geometries because it preserves the dominant thermal behavior while maintaining analytical simplicity compared with fully transient multi-dimensional models, which often require extensive thermophysical and boundary-condition information [11]. Within this framework, heat transfer primarily occurs through convection, conduction, and radiation. Convection governs heat exchange between the product surface and the surrounding environment

and is described by Newton's law of cooling:

$$q_{\text{conv}} = h\Delta T, \tag{1}$$

where h is the convective heat transfer coefficient and ΔT is the temperature difference between the surface and surrounding environment. Internal heat propagation is governed by Fourier's law of conduction, which relates heat flux to the temperature gradient and thermal conductivity:

$$q_{\text{cond}} = -k\frac{dT}{dy}, \tag{2}$$

where k denotes the thermal conductivity of the material and $\frac{dT}{dy}$ represents the spatial temperature gradient. Radiative exchange is not explicitly modeled in the simplified formulation [10], consistent with the reduced-order treatment adopted for the controlled low-temperature experimental configuration. Heat transfer in frozen food is inherently transient, particularly immediately after exposure to a warmer environment, where rapid surface heating creates strong thermal gradients and surface-to-core thermal lag. This transient behavior can be quantitatively described using the Fourier number (Fo), defined as [10]:

$$Fo = \frac{\alpha t}{L^2}, \tag{3}$$

where t is the elapsed time, L is the conduction path length between the heat-exposed surface and the geometric core, and α is the thermal diffusivity. As illustrated in Fig. 2, frozen food transitioning from $-30\,^{\circ}\text{C}$ to ambient conditions undergoes several thermal stages. During the Early Transient stage ($Fo < 0.2$), the Surface Temperature (T_s) rises rapidly while the Core Temperature (T_c) remains relatively unchanged, resulting in pronounced thermal lag.

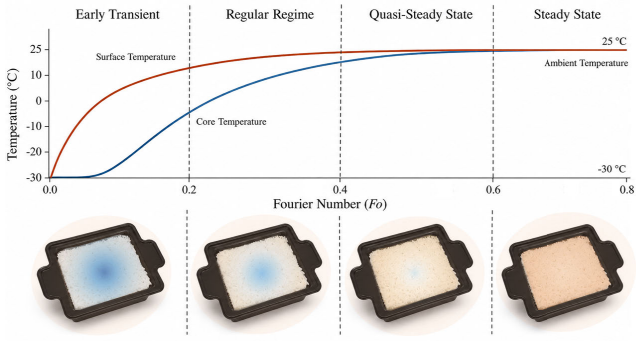


FIGURE 2. Conceptual thermal stages during the reheating process.

Once $Fo \geq 0.2$, the system transitions into a regular regime, where thermal diffusion has propagated across the dominant conduction path and the influence of strongly transient effects is substantially reduced [10], [11]. Within this regime, T_s and T_c evolve in a more consistent and correlated manner [11]. As heating progresses, the thermal response may gradually approach quasi-steady behavior, before ultimately converging toward the Ambient Temperature (T_∞) of 25 °C under steady-state conditions.

Overall, this physically interpretable thermal relationship establishes the theoretical foundation for non-invasive internal temperature estimation. Nevertheless, practical implementation relies heavily on precise and reliable surface temperature acquisition, which can be achieved through advanced sensing and imaging technologies.

C. COMPUTER VISION AND OBJECT SEGMENTATION

Computer vision has become an important technology for automated and non-destructive food quality inspection. For example, hyperspectral imaging is extensively utilized for analyzing chemical distribution [12], while X-ray imaging focuses on foreign body detection. Among various imaging modalities, infrared thermography [13] is widely used for non-contact thermal monitoring by measuring emitted infrared radiation. In frozen food applications, LWIR imaging is particularly effective due to its sensitivity within low-temperature operating ranges [14], [15]. However, thermal images alone generally lack sufficient structural information for reliable object localization in industrial environments.

To improve localization accuracy, multimodal thermography combining RGB and thermal imaging has been increasingly adopted [16]. RGB imagery provides detailed spatial features, while LWIR imaging supplies thermal information. Through multimodal image registration and alignment [17], synchronized RGB-thermal imaging enables accurate thermal ROI extraction. Recent advances in deep learning have introduced specialized architectures for detection tasks across different imaging domains. In hyperspectral imaging, blind-block reconstruction and pixel-shuffle downsampling blind-spot reconstruction networks have been developed for anomaly detection [18], [19], while spectral-prior-guided object detection exploits spectral information for object detection in hyperspectral remote sensing images [20]. Deep learning has also been widely applied to RGB-based agricultural and food-vision tasks [21], [22]. These studies demonstrate that the advantages of different deep learning architectures depend on the characteristics and requirements of the target detection task. For real-time RGB-based food localization in the present framework, the YOLO family [23] provides efficient object detection and instance segmentation suitable for industrial deployment.

D. RELATED WORK

Generally, non-invasive internal temperature estimation has relied on specialized sensing technologies such as ultrasonic methods [8], NIR spectroscopy [4], and MRI/NMR-based techniques [9]. Ultrasonic sensing is influenced by acoustic attenuation and impedance variations, NIR methods require robust models for temperature-gradient and product-dependent spectral variations, and MRI/NMR-based techniques involve specialized instrumentation and complex signal interpretation. Consequently, these specialized sensing approaches remain challenging for continuous, high-throughput conveyor-based inspection systems. These limitations have motivated inferential thermal modeling

approaches, where internal temperatures are estimated from externally measurable thermal information.

Surface-temperature-based internal thermal estimation has been investigated through inverse and physics-guided heat-transfer formulations [24], [25]. Recent thermography-based studies have further extended surface-to-internal thermal inference to granular and particulate media, subsurface defect detection, photothermal reconstruction, and infrared-image-based temperature monitoring [26], [27], [28], [29]. Related approaches have also been extended to biomedical thermal diagnostics [30], [31] and lithium-ion battery thermal management systems [32], [33], [34]. These studies demonstrate the effectiveness of parameter estimation, reduced-order modeling, thermography-based inversion, and observer-based approaches for estimating inaccessible internal thermal conditions. In biomedical and battery systems, sensors are typically fixed on stationary structures, enabling continuous time-series analysis through physics-machine learning hybrids or state observer-based methods [35], [36].

However, these approaches focus on reconstructing internal temperature fields and often require transient thermal sequences, iterative optimization, or continuous data acquisition. Their direct application to industrial food processing is therefore challenging because high-throughput operation [12] and practical sensing constraints [13] limit continuous optical access to moving products. Moreover, infrared thermography measures only surface temperature, while temperature gradients [4], low thermal conductivity, and strong surface-to-core thermal lag complicate internal temperature inference in frozen foods. Accordingly, the present work focuses on estimating only the geometric-center core temperature required for HACCP monitoring from a single surface observation within the regular regime.

Therefore, a critical gap remains in developing a computationally efficient framework that integrates multimodal computer vision, reliable thermal ROI extraction, and physics-guided core temperature estimation under realistic industrial constraints. Rather than performing full thermal reconstruction, the proposed framework applies established heat-transfer principles within a reduced-order one-dimensional formulation to infer the geometric-center core temperature from a single synchronized RGB-LWIR observation. The proposed framework addresses this gap by combining RGB-LWIR multimodal segmentation with a simplified 1D heat transfer model applied within the regular regime. A 3-fold leave-one-batch-out validation strategy is then used to evaluate the consistency of the physics-guided coefficient across independent food samples, enabling accurate non-invasive core temperature estimation from a single instantaneous surface thermal snapshot.

E. EVALUATION METRICS

Performance evaluation in computer vision-based temperature estimation systems typically involves both segmentation and regression metrics. For object detection and segmentation, Precision, Recall, F_1-score, and mean Average Precision (mAP) are commonly adopted to assess localization and classification performance [37]. For temperature estimation, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2) are widely used to quantify estimation accuracy and model reliability [38]. Furthermore, confidence intervals and Bland-Altman evaluation are employed to assess the statistical robustness and measurement agreement between the non-invasive estimates and reference measurements [11], [39].

III. METHODOLOGY

This section describes the proposed non-invasive approach for estimating the core temperature of frozen food, along with the experimental configuration employed for system validation. The methodology combines synchronized RGB and thermal imaging, deep learning-based segmentation, and physics-guided heat transfer analysis to achieve automated and reliable core temperature prediction from surface thermographic measurements. To address the limitations of conventional methods, a lightweight and industrially deployable physics-guided framework is developed for direct estimation of frozen food core temperature from instantaneous surface thermal data. The proposed approach integrates:

- Instantaneous RGB-LWIR multimodal thermography
- ROI-aware thermal feature extraction
- Physics-guided heat transfer modeling within the regular regime
- Single-shot image acquisition at a designated inspection checkpoint
- Non-invasive operation without embedded probes
- Elimination of continuous temporal thermal tracking
- Low computational complexity suitable for real-time industrial monitoring

As illustrated in Fig. 3, the proposed pipeline consists of five main stages: A: Dataset Acquisition and Preprocessing for construction of a synchronized RGB-thermal dataset; B: Object Detection and Segmentation for localization of the container and food regions; C: ROI-aware Thermal Feature Extraction for reliable surface temperature acquisition; D: Physics-guided Core Temperature Estimation using 1D heat transfer model; and E: Performance Evaluation for both segmentation and temperature estimation.

A. DATASET ACQUISITION AND PREPROCESSING

Frozen cooked rice processed using an IQF system was selected as the representative food sample for this study. Rice is one of the world's most widely consumed staple foods and an economically important agricultural commodity, particularly in Asia [40], making it a practical benchmark for frozen food monitoring applications. In addition, frozen cooked rice exhibits relatively homogeneous thermophysical properties and a single-compartment structure, allowing its thermal behavior to be approximated using a single effective thermal conductivity (k). This minimizes the influence of material heterogeneity and provides a controlled environment for validating the proposed framework.

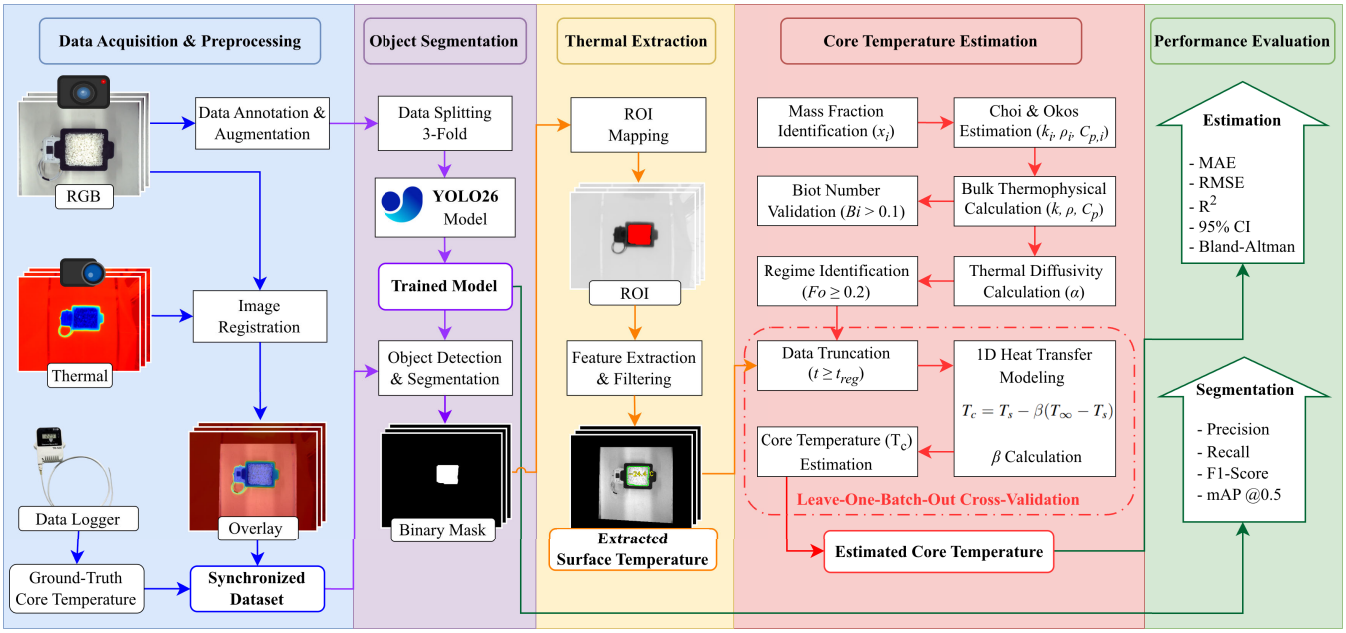


FIGURE 3. Overview of research procedure.

Accordingly, frozen cooked rice was employed in the initial validation stage to evaluate the proposed physics-guided surface-to-core temperature inference mechanism under controlled and interpretable conditions. A simplified thermal model was intentionally adopted to isolate the effectiveness of the proposed methodology before extending it to more structurally complex food products. The uniform composition of frozen cooked rice also facilitates reliable object detection and segmentation.

To obtain ground-truth measurements, a precision temperature logger equipped with an external probe was used to continuously monitor the internal temperature throughout the experiment. To avoid occlusion during thermal imaging, a lateral channel was pre-drilled to the geometric center of each sample, located 15 mm from the exposed surface within the 30 mm total depth, as illustrated in Fig. 4. This enabled horizontal probe insertion without interfering with surface observations. The measured core temperatures were subsequently used as ground-truth references for calibrating the physics-guided coefficient and evaluating estimation performance.

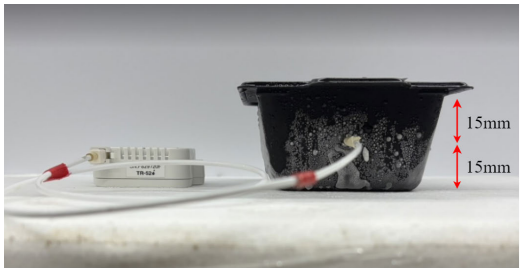


FIGURE 4. Frozen cooked rice sample with an embedded temperature probe.

The experimental analysis focused on a core temperature range between approximately −25 °C and −18 °C, covering both non-compliant warming conditions and overcooling scenarios relevant to IQF energy optimization. Following the IQF process, 45 frozen cooked rice containers collected from three independent IQF production batches were removed directly from the production system and exposed to room temperature for approximately 60 s during transportation and setup before placement on the imaging platform. Each experiment lasted 10 minutes, during which synchronized RGB and thermal images were captured at 3-second intervals, yielding 200 frames per sample and a total dataset of 9,000 multimodal image pairs. Ground-truth core temperature measurements were temporally synchronized with the corresponding image frames using timestamp alignment, ensuring consistency between internal temperature measurements and surface thermal observations.

Fig. 5 illustrates the top-view configuration of the experimental acquisition system. A calibrated dual-camera setup was mounted 40 cm above the sample platform, integrating an industrial RGB camera for spatial localization and a radiometric LWIR camera (8–14 μm, 640 × 512 resolution) for surface temperature acquisition. To support the 1D heat conduction approximation used in the proposed thermal model, samples were placed on expanded polystyrene insulation to minimize conductive heat transfer through the base. Consequently, heat transfer occurred predominantly through the upper exposed surface to a controlled ambient environment maintained at 25 °C.

Due to differences in spatial resolution, lens distortion, and field of view, the RGB and thermal modalities exhibit substantial geometric discrepancies. Therefore, cross-modal calibration was performed to estimate intrinsic and extrinsic

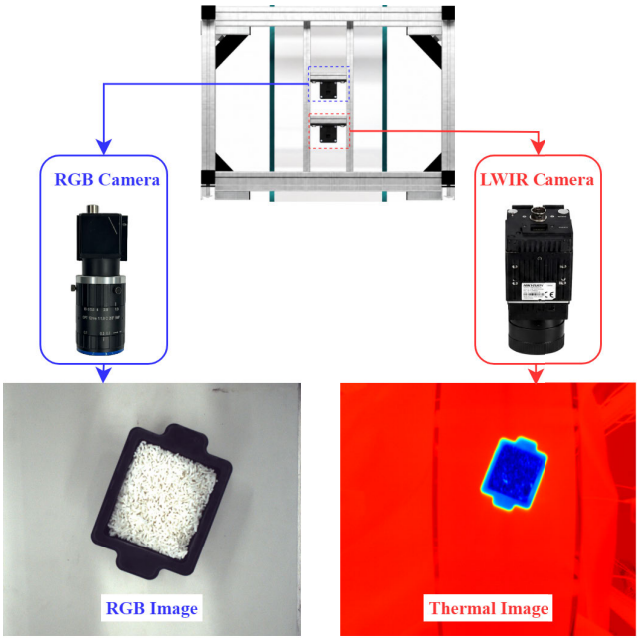


FIGURE 5. Experimental RGB-LWIR acquisition setup.

parameters for accurate spatial alignment between sensors. The multimodal framework combines the high spatial resolution of RGB imaging for object localization with LWIR thermography for surface temperature extraction. To achieve pixel-level correspondence, thermal images were geometrically registered into the RGB coordinate space. This alignment enabled accurate association between detected ROIs and their corresponding thermal signatures, which is essential for downstream temperature estimation. Following registration, supervised annotations were generated on RGB images to produce bounding boxes for object detection training.

To improve robustness and generalization, geometric augmentation techniques including random rotation, scaling, and flipping were applied during training. These augmentations increase variability in object orientation and spatial configuration, thereby improving detection performance under diverse imaging conditions. The synchronized multimodal dataset was subsequently used to validate the proposed infrared thermography framework for non-invasive, real-time surface temperature acquisition and physics-guided core temperature estimation with emissivity correction.

B. OBJECT DETECTION AND SEGMENTATION

Accurate thermal extraction requires precise spatial delineation of food regions while maintaining real-time performance for core-temperature estimation of frozen foods. This second stage implements a YOLO26-seg instance segmentation framework [23] to generate pixel-level masks for multimodal fusion. To rigorously evaluate the model's generalization across independent production batches, a 3-fold Leave-One-Batch-Out (LOBO) cross-validation strategy was adopted, as shown in Fig. 6. In each fold, data from two production batches were used for training, while the remaining batch was reserved exclusively for testing, resulting in a $2:1$ train-test split. This protocol ensured that the model was always evaluated on a completely unseen production batch, thereby preventing information leakage between batches and providing a realistic assessment of robustness under batch-to-batch variability. The model was trained with an input resolution of $2{,}048 \times 2{,}048$ pixels, a batch size of 4, and 100 epochs. Optimization was performed using the Muon stochastic gradient descent optimizer with an initial learning rate of 1×10^{-2}, a linear learning-rate decay schedule, and a weight decay of 5×10^{-4}.

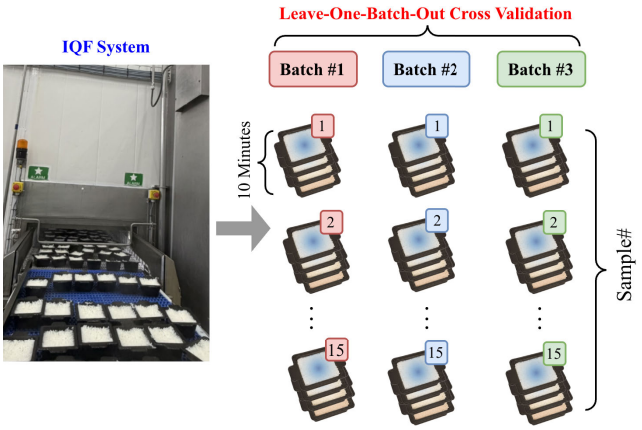


FIGURE 6. IQF production batches used in 3-fold LOBO cross-validation.

The multi-task loss function comprises Intersection over Union (IoU) loss for bounding box regression, binary cross-entropy for classification, and pixel-wise binary cross-entropy for mask prediction. A trained segmentation model is thus obtained and subsequently evaluated on the held-out test set in a blinded manner during the performance evaluation phase. Following precise instance segmentation, the subsequent phase focuses on extracting representative thermal features from the infrared domain. The trained segmentation model generates a binary mask for each detected food instance, where pixels belonging to the food region (ROI) are assigned a value of 1 while background pixels are assigned a value of 0. This pixel-level mask acts as a spatial filter, restricting thermal analysis exclusively to the segmented food region and minimizing background-induced temperature bias.

C. THERMAL EXTRACTION

Thirdly, to ensure that only spatially coherent and thermally representative pixels contribute to surface temperature estimation, visual segmentation masks are mapped to the thermal domain followed by thermal refinement. Since accurate core-temperature inference fundamentally depends on the extracted surface temperature, multimodal segmentation and ROI refinement are not auxiliary preprocessing steps but essential components of the framework. **1) ROI Mapping**: Binary masks generated from RGB-based instance

segmentation are projected onto corresponding registered thermal images. Since geometric calibration and multi-modal registration were established during preprocessing, pixel-level correspondence between modalities is preserved. Consequently, the segmented ROIs are transferred to the thermal domain with minimized ROI-association ambiguity, effectively isolating the food surface while suppressing background regions and container boundaries that may introduce thermal contamination. **2) Thermal Feature Extraction and Anomaly Filtering**: After ROI mapping, surface temperature values are extracted exclusively from the segmented food region. To refine this region, morphological erosion is applied to the binary mask to remove outer boundary pixels, reducing mixed-pixel artifacts at food-background transition regions. Additionally, bulk rice matrices inherently contain porous microstructures and interstitial air pockets [41], [42], which can produce localized thermal anomalies caused by shadowing effects, irregular thermal emission, reflection, or mixed-pixel artifacts. To mitigate these non-uniform thermal signatures, a physically motivated percentile filtering strategy suppresses extreme low- and high-temperature outliers. Finally, the spatial average of all valid thermal pixels within the refined ROI is computed to obtain the extracted surface temperature. Spatial and temporal filtering are applied only to suppress measurement artifacts, including boundary-induced mixed pixels, extreme thermal outliers, and non-physical frame-to-frame fluctuations, while preserving the underlying thermal trend. This provides a stable extraction of the representative surface temperature while minimizing background interference.

D. CORE TEMPERATURE ESTIMATION

To accurately estimate the core temperature of the frozen cooked rice, its bulk thermophysical properties must first be determined. The procedure initiates with Mass Fraction Identification, where the mass fractions (X_i) of the primary chemical constituents of component i—namely moisture, carbohydrate, protein, fat, and ash—are acquired from the USDA Standard Reference Database [43]. Based on these identified constituents, Choi and Okos's Empirical Models [44] are subsequently employed to evaluate the temperature-dependent thermophysical properties of each individual component as a function of the reference temperature (T):

$$P_i(T) = a_i + b_i T + c_i T^2, \tag{4}$$

where $P_i(T)$ represents the specific heat capacity ($C_{p,i}$), density (ρ_i), or thermal conductivity (k_i) of component i as a function of temperature T. The empirical coefficients (a_i, b_i, and c_i) are sourced from established engineering literature [45] and are detailed in Table 1. In this work, T was set to 20 °C as a nominal reference condition for estimating representative bulk thermophysical properties, consistent with the standard reference temperature used in industrial metrology [46]. These values were used for regime identification and dimensionless heat-transfer assessment rather than as exact temperature-resolved properties of the frozen matrix.

TABLE 1. Empirical coefficients for rice constituents.

Component	a	b	c
$C_{p,i}$ (kJ/kg·K)			
Water	4.1762	-9.09×10^{-5}	5.47×10^{-6}
Carbohydrate	1.5488	1.96×10^{-3}	-5.94×10^{-6}
Protein	2.0082	1.21×10^{-3}	-1.31×10^{-6}
Fat	1.9842	1.47×10^{-3}	-4.80×10^{-6}
Ash	1.0926	1.89×10^{-3}	-3.68×10^{-6}
k_i (W · m^{-1} · K^{-1})			
Water	0.5711	1.76×10^{-3}	-6.70×10^{-6}
Carbohydrate	0.2014	1.39×10^{-3}	-4.33×10^{-6}
Protein	0.1788	1.20×10^{-3}	-2.72×10^{-6}
Fat	0.1807	-2.76×10^{-4}	-1.77×10^{-7}
Ash	0.3296	1.40×10^{-3}	-2.91×10^{-6}
ρ_i (kg/m^3)			
Water	997.18	3.14×10^{-3}	-3.76×10^{-3}
Carbohydrate	1599.1	-0.3105	–
Protein	1329.9	-0.5184	–
Fat	925.59	-0.4176	–
Ash	2423.8	-0.2806	–

Subsequently, the obtained individual component properties are integrated based on their respective mass and volume fractions (X_i^v). This step performs the Bulk Thermophysical Calculation. The bulk thermal conductivity (k), density (ρ), and specific heat capacity (C_p) of the composite food matrix are defined by the following additive relationships:

$$k = \sum_i X_i^v k_i, \tag{5}$$

$$\rho = \frac{1}{\sum_i \frac{X_i}{\rho_i}}, \tag{6}$$

$$C_p = \sum_i X_i C_{p,i}. \tag{7}$$

These resulting bulk parameters directly establish the physical foundation required for the subsequent heat-transfer assessment. Utilizing the determined k, the Biot Number Validation is performed to assess the presence of non-negligible internal temperature gradients. Bi is evaluated using the conduction path length (L) of 0.015 m and h according to the standard dimensionless formulation:

$$Bi = \frac{hL}{k}. \tag{8}$$

For the present post-freezing air-exposure condition, h was considered within a representative range of 10 to 50 W · m^{-2} · K^{-1} [45]. The lower-bound value ($h_{\text{worst}} = 10$ W · m^{-2} · K^{-1}) was used as a conservative condition for the Biot Number Validation. The calculated Bi was then compared against the lumped capacitance threshold ($Bi = 0.1$) to assess whether internal temperature gradients could be neglected within the matrix. Radiative exchange was not explicitly modeled as an independent heat-transfer mode in the reduced-order formulation [10], because the

experiment was conducted under low-temperature, single-container, controlled laboratory conditions with minimized surrounding view-factor coupling. Following this validation, the Thermal Diffusivity Calculation is executed using the integrated bulk parameters (k, ρ, and C_p) to determine α, which is defined by:

$$\alpha = \frac{k}{\rho C_p}. \tag{9}$$

This calculated property is utilized rather than relying on previously reported thermal diffusivity values for rice, which ranged from approximately 0.91×10^{-7} to 3.23×10^{-7} m^2/s [41], [42]. Those literature values were obtained under significantly different material conditions and thermal characteristics.

Using the determined α, Regime Identification is performed to define the temporal boundary for excluding the strongly transient initial stage. This boundary is governed by the condition $Fo \geq 0.2$. By applying this threshold as an operational criterion for regular regime selection, the transition time (t_{reg}) was estimated as:

$$t_{\text{reg}} = \frac{0.2L^2}{\alpha}. \tag{10}$$

Accordingly, the Data Truncation step was applied to retain only observations satisfying $t \geq t_{\text{reg}}$, where the assumptions underlying the one-dimensional heat-conduction analysis are valid. It should be emphasized that t_{reg} represents the elapsed thermal evolution time after the product exits the IQF system rather than the residence time at the inspection checkpoint. In practical production lines, frozen products are continuously conveyed from the IQF system to downstream processes, such as lid sealing and packaging, during which they naturally warm under ambient conditions. Therefore, the elapsed time required to reach the regular regime is inherently provided by the existing conveyor transport and is determined by the production-line layout and the location of the inspection checkpoint. Consequently, the proposed framework requires only a single synchronized RGB-LWIR image pair to be acquired at the designated inspection checkpoint, without continuous thermal monitoring or repeated image acquisition. Within the truncated temporal domain, a reduced-order physics-guided surface-to-core heat-transfer relationship is established by applying the one-dimensional heat-conduction approximation under the regular-regime assumption [10]. This relationship links the convective heat flux entering the exposed material surface from the ambient environment (q_{conv}) with the internal conductive heat flux (q_{cond}) propagating toward the interior, which is expressed as follows:

$$h(T_\infty - T_s) = \frac{k(T_s - T_c)}{L}. \tag{11}$$

By rearranging this relationship, T_c can be expressed directly as a function of the extracted T_s. In practical industrial environments, continuously tracking process-dependent and specimen-specific parameters such as h, k, and L is unfeasible. Therefore, the combined influence of these heat-transfer parameters is consolidated and represented by a single physics-guided coefficient, denoted as β, yielding the following 1D Heat Transfer Model:

$$T_c = T_s - \beta(T_\infty - T_s). \tag{12}$$

To determine the value of β from experimental observations while preventing data leakage, a 3-fold LOBO cross-validation strategy was employed. In each fold, two production batches (30 samples) were used for coefficient determination, while the remaining production batch (15 samples) was reserved exclusively for performance evaluation. Within each coefficient-determination fold, β was first calculated at every valid observation using the rearranged form of (12). These values were subsequently averaged to obtain a sample-specific β, and the mean across the 30 determination samples was adopted as the generalized coefficient for that fold. The resulting coefficient was then applied without further adjustment to estimate the core temperatures of the unseen testing batch. This partitioning ensured that each production batch served as an independent testing set exactly once, providing a rigorous assessment of the coefficient's generalizability across production batches.

Ultimately, the final Core Temperature Estimation is achieved by substituting the generalized physics-guided coefficient β into the proposed 1D heat transfer model. The resulting framework enables non-invasive estimation of T_c from an instantaneous observation of T_s, provided that the analysis is restricted to the validated regular thermal regime.

E. PERFORMANCE EVALUATION

The proposed framework is evaluated in terms of both segmentation and temperature estimation performance. The vision module is assessed using Precision, Recall, F_1-score, and mAP@0.5. Core temperature estimation accuracy is evaluated using MAE, RMSE, and R^2 against probe-based ground-truth measurements. To assess statistical robustness, 95% confidence intervals are obtained through 3-fold LOBO cross-validation, while uncertainty propagation analysis is conducted to examine the influence of variations in key thermophysical and measurement parameters. Finally, Bland-Altman analysis is performed to quantify systematic bias and the 95% Limits of Agreement (LoA) between the estimated and measured core temperatures.

IV. RESULTS AND DISCUSSION

Following extensive theoretical derivation, thermophysical characterization, and experimental validation, this section presents the performance evaluation of the proposed non-invasive core temperature estimation framework. The analysis focuses on both the object detection and instance segmentation performance, as well as the accuracy of the proposed core temperature estimation approach.

A. DATASET ACQUISITION AND PREPROCESSING OUTCOMES

The dataset acquisition and preprocessing stage successfully established a high-quality multimodal dataset for subsequent model development and evaluation. A total of 9,000 synchronized RGB-thermal image pairs were acquired from 45 frozen cooked rice samples collected across three independent IQF production batches, covering the target core temperature range of $-25\,^{\circ}\mathrm{C}$ to $-18\,^{\circ}\mathrm{C}$, with each sample contributing 200 temporally aligned frames over a 10-minute observation period. Robust temporal synchronization and spatial alignment between RGB and thermal modalities were achieved through timestamp-based frame pairing and image registration.

As illustrated in Fig. 7, the registration process effectively corrected positional offsets and geometric distortions caused by adjacent camera placement, resulting in accurate cross-modality correspondence suitable for reliable multimodal fusion and thermal analysis. To further improve model robustness and generalization, data augmentation techniques including rotation, mosaic, and brightness adjustment were applied to the training set. These transformations simulated variations in camera orientation and illumination conditions while preserving the raw thermal intensity distribution within each object ROI.

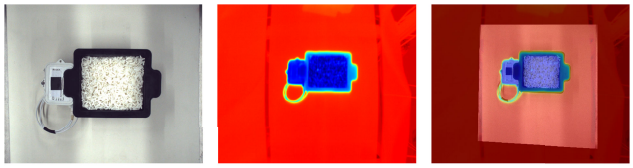


FIGURE 7. Image registration process.

B. DETECTION AND SEGMENTATION PERFORMANCE

Building upon the preprocessed high-resolution RGB images, the YOLO26-seg architecture was employed to isolate the target food regions from the surrounding packaging. As shown in Fig. 8, the model successfully delineated complex object boundaries with high spatial precision, enabling accurate extraction of relevant thermal regions for subsequent analysis.

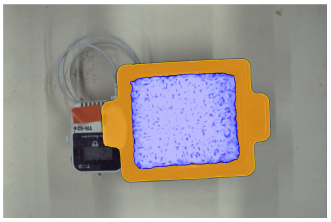


FIGURE 8. Detection and segmentation of the frozen cooked rice sample.

At an IoU threshold of 0.5, the trained model achieved a Precision of 0.9613, Recall of 0.9342, F_1-score of 0.9476, and mAP@0.5 of 0.9779. These results demonstrate the robustness of the proposed segmentation framework in minimizing background interference while preserving accurate food surface boundaries. Such spatial precision is critical for reliable multimodal thermal analysis and subsequent core temperature estimation in industrial monitoring environments.

C. SURFACE THERMAL CHARACTERIZATION

The multimodal image registration framework successfully aligned the RGB-derived segmentation masks with the corresponding thermal frames. As illustrated in Fig. 9, the YOLO-based segmentation output was converted into a binary mask to isolate the frozen cooked rice regions. To reduce thermal boundary interference from the experimental platform and packaging tray, morphological erosion was applied prior to thermal extraction. This operation contracted the mask boundaries inward, excluding edge pixels potentially affected by external thermal artifacts and producing a more reliable target-specific thermal ROI.

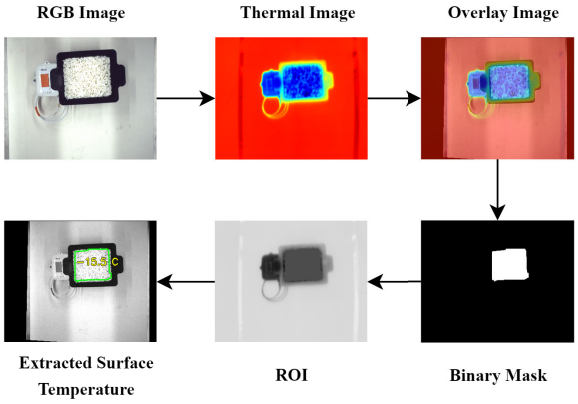


FIGURE 9. Thermal ROI extraction after multimodal registration and mask erosion.

Following ROI extraction, the relationship between the extracted surface temperature and the measured core temperature of each frame was analyzed to assess the feasibility of estimating internal temperature using surface thermography. Fig. 10 presents the measured core temperatures and the corresponding extracted surface temperatures for all 45 containers collected from three independent IQF production batches at $t > 60\,\mathrm{s}$ after removal from the IQF system. The surface temperature increased rapidly due to direct exposure to ambient conditions, whereas the core temperature exhibited a slower and more gradual rise governed by transient heat conduction within the rice matrix. This delayed response is attributed to the relatively low thermal conductivity of the sample, where the outer layers temporarily act as a thermal barrier before heat propagates toward the geometric center. Despite the existence of a transient thermal gradient between the surface and core regions, the consistent thermal behavior observed across

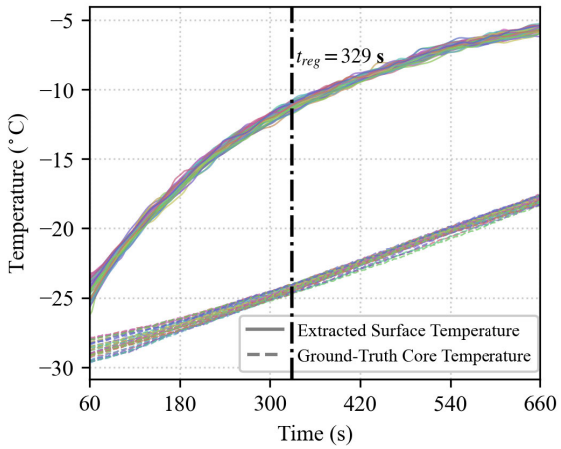


FIGURE 10. Surface and core temperature profiles during ambient exposure (t > 60 s).

all samples supports the feasibility of non-invasive core temperature estimation using surface thermography.

Fig. 11 further illustrates the relationship between the extracted surface temperatures and the experimentally measured ground-truth core temperatures across all 45 samples. The strong relationship observed between the extracted surface temperature and the measured core temperature demonstrates that the externally observable thermal response retains substantial information regarding the otherwise inaccessible internal core temperature. This observation establishes the physical basis for non-invasive core temperature inference from instantaneous thermal measurements.

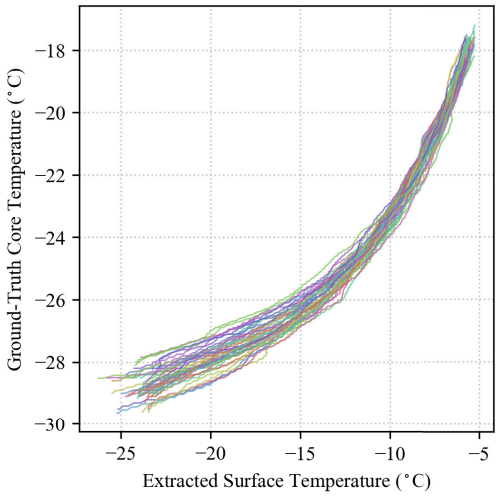


FIGURE 11. Demonstration of internal thermal-state observability from instantaneous surface measurements.

D. CORE TEMPERATURE ESTIMATION ANALYSIS

Following the workflow for the core temperature estimation outlined in Fig. 3, Fig. 12 presents the intermediate results of the proposed core temperature estimation framework.

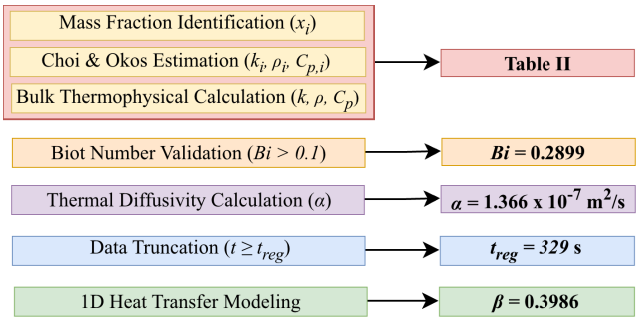


FIGURE 12. Intermediate results of the proposed core temperature estimation procedure.

First, Table 2 presents the thermophysical properties used for core temperature estimation. The constituent properties were first calculated at the reference temperature of 20 °C using (4), after which the effective bulk thermal conductivity (k), density (ρ), and specific heat capacity (C_p) were obtained using the additive relationships defined in (5)–(7).

TABLE 2. Constituent mass fractions and the resulting bulk values of the matrix.

Mass Fraction (X_i)	$C_{p,i}$ (kJ/kg·K)	k_i (W · m^{-1} · K^{-1})	ρ_i (kg/m^3)
Water (0.684)	4.1766	0.6036	995.74
Carbohydrate (0.282)	1.5856	0.2274	1592.89
Protein (0.027)	2.0318	0.2016	1319.53
Fat (0.003)	2.0117	0.1751	917.24
Ash (0.004)	1.1289	0.3564	2418.19
Bulk Property	**3.3693**	**0.5175**	**1124.42**

Utilizing these established bulk properties, the analysis yielded $Bi = 0.2899$ under the previously defined worst-case condition ($h_{\text{worst}} = 10\,\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$) via (8). This value explicitly exceeds the lumped capacitance threshold ($Bi > 0.1$), thereby indicating that internal temperature gradients cannot be neglected within the matrix. Using the bulk properties from Table 2, the thermal diffusivity α was calculated according to (9) to be 1.366×10^{-7} m^2/s. Applying this diffusivity to the regular-regime threshold ($Fo \geq 0.2$), t_{reg} was explicitly evaluated via (10), identifying the transition at 329 s. This threshold marks the point at which thermal diffusion has propagated across the defined conduction path length and the influence of strongly transient thermal behavior is substantially reduced. Consequently, subsequent analyses were restricted to this regular thermal regime ($t \geq 329$ s), where the physics-guided coefficient β governing the core temperature estimation in (12) was evaluated to be 0.3986. The package-specific coefficient estimates yielded a mean value of 0.3986 with a 95% confidence interval of ± 0.0024 across 45 independent packages. This narrow parameter distribution indicates stable coefficient estimation across the investigated packages under the controlled experimental conditions and supports its suitability for core temperature estimation within this specific food matrix.

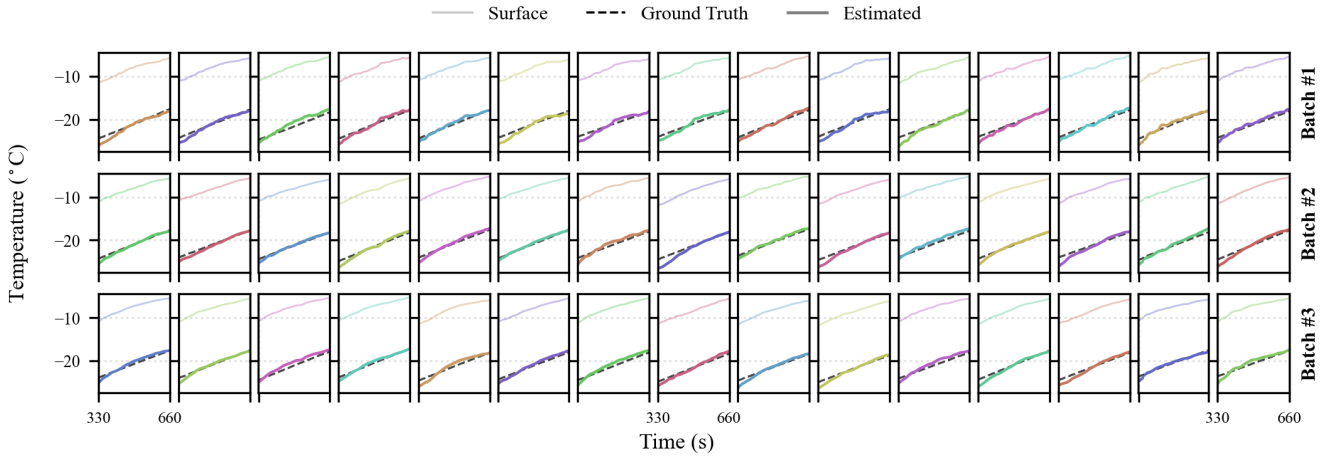


FIGURE 13. Extracted surface temperature along with estimated core temperature versus ground-truth core temperatures for all samples at $t \geq t_{\text{reg}}$.

Although β originates from the analytical heat-transfer formulation, it should be interpreted as an effective physics-guided parameter within the simplified 1-D model. It captures residual effects of geometric simplifications, boundary-condition uncertainty, emissivity-related surface-temperature uncertainty, thermal heterogeneity, and measurement imperfections that are not explicitly represented in the reduced-order formulation. Accordingly, β is configuration-specific and should not be assumed to be universally transferable across different food products or processing conditions. Variations in ambient temperature within the validated operating range can be accounted for through the explicit ambient-temperature term in (12) without necessarily requiring recalibration. However, changes in food composition, product geometry, packaging configuration, or airflow conditions that substantially alter the effective heat-transfer characteristics require recalibration of β using representative measurements from the corresponding configuration.

The sample-specific performance of the proposed physics-guided estimation framework across all 45 independent validation cases is presented in Fig. 13. The estimated core temperatures exhibit strong agreement with the probe-based ground-truth measurements throughout the evaluated thermal range, demonstrating the capability of the framework to estimate internal core temperature from externally observable surface temperatures. Fig. 14 further provides experimental evidence that the observable surface thermal response retains sufficient information to infer the hidden core temperature, thereby supporting the practical observability of core temperature from surface thermography.

A notable observation is the presence of a brief spatiotemporal lag immediately after the regular-regime threshold ($t_{\text{reg}} = 329\,\text{s}$). Although the Fourier criterion ($Fo \geq 0.2$) indicates the onset of the regular regime, it does not imply the immediate elimination of higher-order transient effects. Residual thermal disturbances therefore persist during this transition period, producing a slight underestimation of the

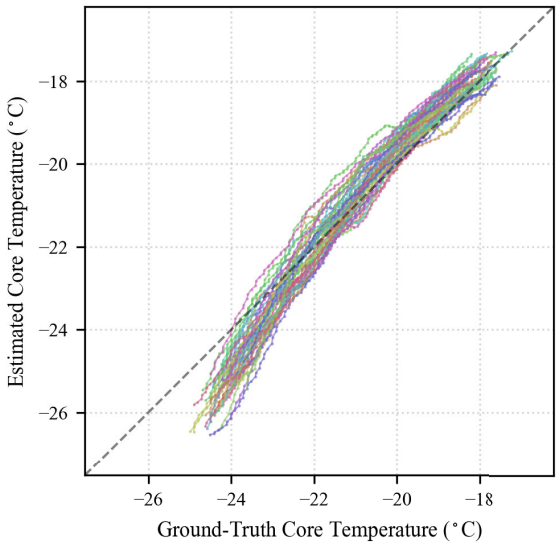


FIGURE 14. Estimated versus measured core temperature.

core temperature. As these transient components decay, the estimated and measured temperature trajectories progressively converge. This behavior supports the use of the regular-regime criterion to exclude the strongly transient initial period, after which the simplified 1D conduction model becomes increasingly applicable.

The observed convergence supports the effectiveness of integrating multimodal thermography, thermophysical characterization, and heat-transfer theory within a unified estimation framework. Rather than relying on empirical correlations or purely data-driven regression models, the proposed methodology establishes a physically interpretable relationship between surface thermal observations and the internal core temperature. Quantitative performance metrics for the finalized core temperature estimation model, summarized in Table 3, further support this observation,

TABLE 3. Quantitative performance metrics and bland-altman analysis.

Performance Metric	Value
MAE (°C, 95% CI)	0.4553 ± 0.0386
RMSE (°C)	0.5814
R^2	0.9050
Bland-Altman Analysis	
Mean Bias (°C)	-0.0218
95% LoA (°C)	$[-1.1266, 1.0830]$

while the Bland-Altman analysis demonstrates minimal systematic bias and narrow limits of agreement, indicating consistent estimation performance across the evaluated core-temperature range.

To further assess methodological robustness, an uncertainty propagation analysis was performed by introducing perturbations in characteristic thickness ($L \pm 10\%$), thermal conductivity ($k \pm 10\%$), ambient temperature ($T_\infty \pm 2\,^\circ\mathrm{C}$), and extracted surface temperature ($T_s \pm 2\,^\circ\mathrm{C}$). The T_s perturbation was intended to represent surface-temperature measurement uncertainty, including residual emissivity-setting effects, radiometric sensor noise, and ROI extraction variability. In this study, the frozen cooked rice samples were acquired directly from an industrial IQF process and remained below $0\,^\circ\mathrm{C}$ throughout the analyzed temperature range (approximately $-25\,^\circ\mathrm{C}$ to $-18\,^\circ\mathrm{C}$), where surface condensation is not expected. Accordingly, the surface emissivity was assumed to remain constant during the experiments. Any residual emissivity variations arising from frost formation or other surface effects were incorporated into the surface-temperature uncertainty represented by the T_s perturbation in the sensitivity analysis. For perturbations affecting L and k, the Biot number and corresponding t_{reg} were recalculated to preserve compliance with the regular-regime boundary condition. The resulting sensitivity profile is presented in Fig. 15.

The analysis shows that the framework is relatively robust to ambient temperature variations and moderate thermophysical-property uncertainties, indicating that the underlying heat-transfer formulation remains stable under realistic operating conditions. For the thermophysical parameters, perturbations in k and L produced non-symmetric changes in MAE because they dynamically shifted the t_{reg} threshold. Specifically, variations that delayed t_{reg} excluded early-phase data with higher residual transient variance, reducing the overall MAE. Conversely, variations that advanced t_{reg} caused the model-applicable period to include more strongly transient data, increasing the error. In contrast, estimation accuracy is most strongly influenced by the extracted surface temperature. Perturbations in T_s directly affected the estimated core temperature, highlighting the critical role of accurate thermal sensing in estimating the internal core temperature. This behavior is consistent with the theoretical formulation, where surface temperature serves as the primary observable boundary condition driving the heat-transfer model.

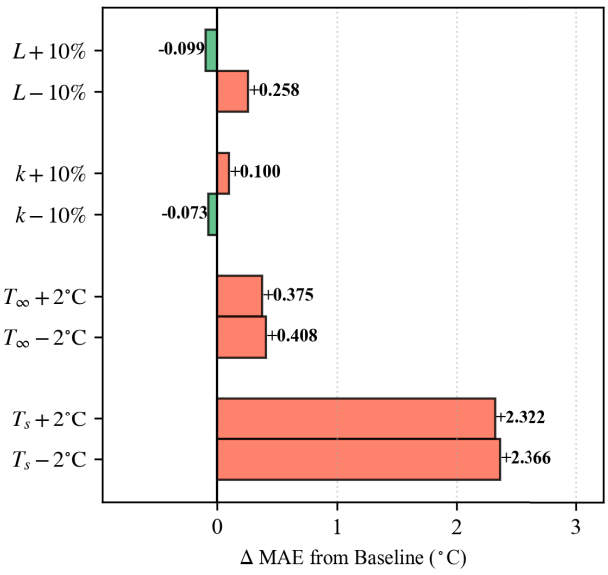


FIGURE 15. Tornado chart of estimation error (MAE) sensitivity to input and property inaccuracies.

These findings reinforce the central premise of the proposed framework: reliable core temperature estimation can be achieved from a single surface thermal observation, provided that the surface thermal boundary information is accurately captured and the thermophysical characteristics of the product are reasonably characterized.

E. DISCUSSION AND LIMITATIONS

The primary contribution of this work lies in establishing a practical methodology for non-invasive core temperature estimation through the integration of multimodal thermography and physics-guided heat-transfer modeling. Beyond achieving accurate temperature estimation, the proposed framework demonstrates how T_s can be systematically transformed into reliable estimates of internal core temperature using thermophysical characterization and conduction theory. By combining RGB-based food localization, LWIR-based surface temperature extraction, and physics-guided thermal analysis within a unified framework, the proposed approach effectively bridges the gap between observable surface conditions and otherwise inaccessible core temperatures.

The experimental results confirm the physical feasibility of the proposed methodology under controlled laboratory conditions. In particular, the observed convergence between estimated and measured temperatures after excluding the strongly transient thermal period supports the applicability of the adopted heat-transfer analysis and demonstrates that internal core temperature can be inferred from a single instantaneous surface thermal observation without invasive probes or continuous thermal monitoring. From an implementation perspective, the proposed framework is intended for deployment at a designated inspection checkpoint

downstream of the IQF system rather than immediately at its outlet. Whether products reach the regular regime before inspection depends on the available thermal-evolution time, which is determined by conveyor speed, transport distance, and production-line layout. Thus, the inspection location should be configured according to the operating conditions of each production line. For the investigated product configuration, the elapsed thermal-evolution time should be at least $t_{\mathrm{reg}} = 329$ s before inspection. For other products or production lines, the corresponding regular-regime threshold should be determined or verified for the target configuration when selecting the inspection point. This requirement does not imply that products must be held stationary or monitored continuously before inspection; rather, the proposed framework estimates core temperature from a single synchronized RGB–LWIR image pair acquired when the product reaches the designated checkpoint. In this context, the term single-shot refers to the use of a single image acquisition for temperature estimation rather than continuous image capture or time-series monitoring at the inspection location.

The uncertainty analysis further reveals that the framework is most sensitive to surface temperature extraction, while remaining comparatively robust to ambient temperature variations and moderate thermophysical-property uncertainties. This highlights the importance of accurate thermal sensing and multimodal image registration, as residual alignment errors, thermal lens distortions, and surface emissivity variations may introduce localized temperature extraction inaccuracies, particularly near object boundaries. The present framework adopts a reduced-order one-dimensional heat-conduction approximation to estimate the geometric-center core temperature rather than reconstructing the complete internal temperature field. Accordingly, the reference temperature measurements in this study were obtained at the geometric center of the frozen samples, consistent with HACCP monitoring practice, while ROI erosion was employed to minimize the influence of boundary-related thermal effects. Experimental validation using multiple internal probe locations to further characterize multidimensional heat-transfer behavior remains an important direction for future work, particularly for finite geometries and heterogeneous frozen foods.

A key limitation of the current study is its reliance on a controlled thermal environment and the assumption that the food product has progressed beyond the initial transient regime. Although the proposed framework was validated using 45 frozen cooked rice specimens collected from three independent IQF production batches, several limitations remain. The current methodology assumes that products have reached the regular thermal regime before image acquisition and was evaluated under controlled laboratory conditions with fixed imaging geometry and environmental conditions. Deployment in practical IQF conveyor systems may introduce additional factors, including product motion, variable camera-to-object distance, product orientation, conveyor speed, and dynamic convective heat-transfer conditions. These factors may affect image registration, thermal measurement stability, and heat-transfer modeling accuracy. Future work will therefore focus on conveyor-based validation, adaptive RGB-LWIR calibration, dynamic thermal-condition characterization, and estimation approaches that relax the regular-regime requirement to improve robustness and deployment flexibility across production lines with different conveyor layouts and transport times.

Overall, the results demonstrate the potential of multimodal thermography combined with physics-guided thermal modeling as a computationally efficient, non-destructive, and scalable alternative to conventional invasive temperature measurement techniques for automated food safety monitoring and intelligent quality control.

V. CONCLUSION AND FUTURE WORK

This study presented a lightweight, physics-guided multimodal thermography framework for non-invasive core temperature estimation in frozen food using synchronized RGB and LWIR imaging. By integrating computer vision with established heat-transfer principles, the framework successfully inferred internal core temperatures from instantaneous surface measurements, requiring invasive probe measurements only during the initial calibration stage for a given product line rather than for routine inspection. The combination of ROI-aware thermal extraction and physics-guided heat transfer modeling within the regular regime ensured stable, low-complexity thermal analysis suitable for industrial deployment. More fundamentally, this study demonstrates that core temperature remains observable from instantaneous surface thermal measurements after excluding the strongly transient initial period. This finding establishes the physical foundation for non-invasive core temperature estimation in frozen food monitoring applications.

Experimental results demonstrated the feasibility of real-time thermal monitoring and automated food safety assessment. Validation using 45 frozen cooked rice samples collected from three independent IQF production batches further demonstrated the robustness of the proposed methodology under batch-to-batch production variability. Beyond strong estimation performance, the framework supports the non-invasive detection of both insufficient freezing and excessive overcooling, shifting conventional threshold compliance toward enhanced quality control and energy-efficient processing.

Despite its effectiveness, the framework remains sensitive to multimodal registration accuracy and thermal lens distortion near peripheral regions. Future work will focus on enhancing RGB-LWIR alignment through advanced distortion correction, extending the framework to heterogeneous multi-component frozen foods through product-specific calibration, investigating estimation approaches that relax the regular-regime requirement, and optimizing the setup for high-speed industrial conveyor systems to enable scalable, intelligent thermal inspection.

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